

# Alma Meili Ren

(860) 754-7288 | arr254@cornell.edu | [Portfolio](#) | [LinkedIn](#)

## EDUCATION

### Cornell University College of Engineering

*Bachelor of Science in Mechanical Engineering*

**Ithaca, NY**

*Aug 2023 – Dec 2027*

**Cumulative GPA:** 3.79

**Relevant Coursework:** Mechanical Design, Mechatronics, Fluid Mechanics, Heat Transfer, Thermodynamics, Dynamics, Mechanics of Materials, System Dynamics, Statics, Energy Storage, Oscillations, Quantum Physics, Electromagnetism.

## TECHNICAL SKILLS

**CAD:** SOLIDWORKS, Autodesk Fusion, Tinkercad.

**Simulation & Analysis:** MATLAB, Simulink, Python, Java, ANSYS.

**Instrumentation & Testing:** LabVIEW, Constant-Temperature Anemometry (CTA), Motion Capture Systems, Oscilloscopes.

**Manufacturing & Prototyping:** FDM 3D Printing, Laser Cutting, Manual Mill, Lathe, Welding, Bandsaw, Metal Fabrication.

## EXPERIENCE

### Reykjavik University Iceland School of Energy

*Energy Lab Intern*

**Reykjavik, Iceland**

*Jun 2026 – Aug 2026*

- Independently conducted wind tunnel test campaigns using CTA, active-grid turbulence generators, and velocity time-series data analysis in Python to investigate wind turbine performance under varying turbulent inflow conditions.
- Eliminated data acquisition variability and reduced setup time by 15 minutes by designing a custom hinged probe-mounting device in Autodesk Fusion using print-in-place techniques with a 0.25 mm joint clearance.

### AguaClara Cornell Engineering Project Team

*Full Team Lead*

**Ithaca, NY**

*Sep 2024 – Jun 2026*

- Directed R&D initiatives of a 50-member engineering team, overseeing project planning, testing, and cross-team collaboration for water treatment technologies deployed in 26 facilities across 4 countries.
- Led the design of the first-ever successful field deployment of a hydraulic ram pump in Honduras after 20 years of development, installing 2 functional units that transport water 7 m vertically at 28 mL/s without electricity.
- Analyzed 525,000+ data points in MATLAB to generate performance curves and identify optimal pump configurations.
- Diagnosed recurring pump failures through root cause analysis and controlled variable testing, isolating failure modes to inform targeted design improvements.
- Developed a nonlinear flat spring system using MASTAN2 and Autodesk Fusion to increase restoring force near valve closure, resolving actuation failures caused by hydrostatic overloading.
- Authored 7 reports, manuals, and posters documenting pump design and performance for open-source hydraulic projects.

### Cornell University Department of Mechanical and Aerospace Engineering

*Teaching Assistant – Dynamics*

**Ithaca, NY**

*Aug 2025 – May 2026*

- Supported 200+ students in building MATLAB models of dynamic systems using state-space methods and ODE solvers.
- Facilitated motion-capture projects where students experimentally validated pendulum models and determined inertial properties of complex satellite-like structures through multi-axis swing tests.

## PROJECTS

### Autonomous Block-Collection Robot – Mechatronics Design Competition

*Mar 2026 – May 2026*

- Integrated sensors, servos, motor drivers, and power systems with an Arduino Uno to enable fully autonomous block-collection operation.
- Designed and prototyped a servo-actuated collection arm in Autodesk Fusion with a scooping mechanism and interlocking joints for laser-cut acrylic assembly.

### SwishStick – Open Design Product Pitch

*Jan 2025 – May 2025*

- Collaborated on end-to-end development of a multi-tool water bottle brush with 4 cleaning modes, spanning user research, mechanical design, prototyping, and a final product pitch.
- Modeled and prototyped 10 custom components in Autodesk Fusion with 0.2 mm tolerance interfaces for precise fits.
- Engineered a rotating detent locking mechanism using a notched ball interface to secure attachments at 2 different angles.
- Doubled product durability through iterative redesign and material substitution by designing and executing performance tests across 5 quality metrics, including incremental drop tests.